EC-201: Engineering Project Management

LECTURE NO 6: *Project Conception and Project Feasibility*

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6.1 Project Conception

- Conception of an Industrial Project is the initial step in the process of defining the actual scope of a project. Project conception generally starts with an indication of a requirement or an opportunity that will benefit the corporate interests.

6.2 Stages of Project Conception (1/2)

- Initial conceptualization of a project has various degrees of complexity, depending on the nature of the specific project and the particular analysis and approval procedures used by a company.
- The conceptual stage involves the following activities:

1. ***Definition of the Requirement or Opportunity-*** The continuity of efficient operations and the opening of the new business areas are the main drives for capital investments for industrial firms. Investment opportunities are detected through operational analysis of current performance and by forecasts of the most likely future scenarios.
2. ***Preliminary Formulation of the Alternatives-*** Project conception continues with development of alternatives capable of fulfilling the expressed objectives. During this phase, the company calls upon the experience and the creativity of its technicians, manager and directors to

generate an adequate group of alternatives to fulfil the expressed needs.

6.2 Stages of Project Conception (2/2)

3. ***Initial Selection of Alternatives-*** After the alternatives have been identified, comparative analyses are made in order to select the most beneficial and to reject the least attractive. The selection process employs a basic *feasibility analysis* of each alternative that will allow the identification of the most attractive options.

6.3 Feasibility Analysis

- A feasibility study is a systematic tool used during the project planning process, shows how a business would operate under a specific stated set of assumptions. These assumptions include the technology used (the facilities, types of equipment, manufacturing process, etc.) and the financial aspects of the project (capital needs, volume, cost of goods, wages etc.).

6.4 What is Feasibility Assessment? (1/6)

- As the name implies, a feasibility study is an analysis of the viability of an idea. The feasibility study focuses on helping answer the essential question of *"should we proceed with the proposed project idea?"* All activities of the feasibility study are directed toward helping answer this question.
- Feasibility studies can be used in many ways but *primarily focus on proposed business ventures*. Farmers and others with a business idea should conduct a feasibility study to determine the viability of their idea before proceeding with the development of the business. Determining early on that a business idea will not work saves time, money and heartache later.

6.4 What is Feasibility Assessment? (2/6)

- *A feasible business venture is one where the business will generate adequate cash flow and profits, withstand the risks it will encounter, remain viable in the long-term and meet the goals of the founders*. The venture can be a new start-up business, the purchase of an existing business, an expansion of current business operations or a new enterprise for an existing business.
- A feasibility study is only one step in the business idea assessment and business development process.
- *A feasibility study is usually conducted after producers have discussed a series of business ideas or scenarios. The feasibility study helps to **"frame" and "elaborate"** specific business alternatives so they can be studied in-depth. During this process the number of business alternatives under consideration is usually quickly*

6.4 What is Feasibility Assessment? (3/6)

- *A market assessment may be conducted to help determine the viability of a proposed product in the marketplace. The market assessment will help you identify opportunities in a market. If no opportunities are found, there may be no reason to proceed with a feasibility study. If opportunities are found, the market assessment can give focus and direction to the construction of business alternatives*.
- The conclusions of the feasibility study should outline in depth the various alternatives examined and the suggestions and strengths and weaknesses of each.
- *The feasibility study will help you assess the trade-off between the risks and rewards of moving forward with the business project*.

6.4 What is Feasibility Assessment? (4/6)

- *The go/no-go decision is one of the most critical in business development. It is the point of no return. Once you have definitely decided to pursue a business venture, there is usually no turning back. The feasibility study will be a major information source in making this decision. This indicates the importance of a properly developed feasibility study*.
- The feasibility study outlines and analyses several alternatives or methods of achieving business success. So, *the feasibility study helps to narrow the scope of the project to identify the best business model*. *The business plan deals with only one alternative or model*. The feasibility study helps to narrow the scope of the project to identify and define two or three scenarios or alternatives.

6.4 What is Feasibility Assessment? (5/6)

- *The feasibility study is conducted before the business plan. A business plan is prepared only after the business venture has been considered or believed to be feasible. The business plan provides the* ***"master plan"*** *for project implementation. If the venture is considered not to be feasible, efforts may be made to correct its deficiencies, other alternatives may be explored, or the idea is dropped.*
- *Project leaders may find themselves under pressure to skip the "feasibility analysis" step and go directly to building a business.*

6.4 What is Feasibility Assessment? (6/6)

- *Important reasons given for not doing feasibility analysis include*:
  - We know it is feasible. An existing business is already doing it.
  - Why do another feasibility study when one was done just a few years ago?
  - Feasibility studies are just a way for consultants to make money.
  - Why not just hire a general manager who can do the study?
  - Feasibility studies are a waste of time. We need to buy the building, tie up the site and bid on the equipment.

6.5 Types of Feasibility (1/4)

Feasibility is of the following types:

1. ***Technical Feasibility-*** This area reviews the *engineering feasibility of the project*, including structural, civil and other relevant engineering aspects necessitated by the project design.
2. ***Managerial Feasibility-*** Demonstrated *management capability*, *availability*, *employee involvement*, *and commitment* are key elements required to find out managerial feasibility.
3. ***Economic Feasibility-*** This involves the feasibility of the proposed project to generate economic benefits. A cost- benefit-analysis (CBA) is required. *CBA is a process that is used to estimate the costs and benefits of decisions in order to fin d the most cost effective alternative or path*

6.5 Types of Feasibility

4. ***Financial Feasibility-*** Financial feasibility involves the capability of the project organization to raise the appropriate funds needed to implement the proposed project.
5. ***Cultural Feasibility-*** Cultural feasibility deals with the compatibility of the proposed project with the cultural environment of the project. For example, religious beliefs may influence what an individual is willing to do or not do.
6. ***Social Feasibility-*** Social feasibility addresses the influences that a proposed project may have on the social system in the project environment. The social structure may be such that certain categories of workers may be i n short supply or non existent.

6.5 Types of Feasibility

(3) 7. ***Safety Feasibility-*** Safety feasibility is another important aspect that should be considered in project planning. Safety feasibility refers to an analysis of whether the proposed project is capable of being implemented and operated safely with minimal adverse effects on the environment.

8. ***Political Feasibility-*** Political considerations often dictate directions for a proposed project. This is particularly true for large projects with significant visibility that may have significant government inputs and political suggestions. For Example: CPEC Project, Pakistan and China Joint Project.

9. ***Environmental Feasibility-*** This study assesses viability of a proposed project from environment perspective, identifying potential issues and threats to the successful completion of the proposed project.

6.5 Types of Feasibility (4/4)

10. ***Market Feasibility-*** The market needs analysis to view the potential impacts of market demand, competitive activities, etc. and market share available.

6.6 Tangible and Intangible Benefits

- Estimating benefits and costs in a timely manner is very difficult. Benefits are often defined as:
  - ***Tangible benefits-*** for which dollarsmay be reasonably quantified and measured.
  - ***Intangible benefits-*** that may be quantified in units other than dollars or may be identified and described distinctive.

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